- Given m_k and ξ_k, find the optimal ϕ_{k+1}, by solving

  $$\max_{\phi} \quad -\langle \nabla_{t,x}\phi, m_k\rangle + G(\phi) - \frac{\tau}{2}||\xi_k - \nabla_{t,x}\phi||^2,$$

  which amounts to minimizing a quadratic problem in $\nabla_{t,x}\phi$. The solution can be found as the solution of a Laplace equation $\tau\Delta_{t,x}\phi = \nabla\cdot(\tau\xi_k - m_k)$, with a space-time Laplacian, with Neumann boundary conditions. These conditions are homogeneous in space, and non-homogeneous, due to the role of G, on $t = 0$ and $t = 1$. Most Laplace solvers can find this solution in time $O(N\log N)$, where N is the number of points in the discretization.

- Given m_k and ϕ_{k+1}, find the optimal ξ_{k+1}, by solving

  $$\max_{\xi\in K_q} \quad \langle \xi, m_k\rangle - \frac{\tau}{2}|\xi - \nabla_{t,x}\phi_{k+1}|^2.$$

  By expanding the square, we see that this problem is equivalent to the projection of $\nabla_{t,x}\phi_{k+1} + \frac{1}{\tau}m_k$, and no gradient appears in the minimization. This means that the minimization may be performed pointwisely, by selecting for each (t,x) the point $\xi = (a,b)$ which is the closest to $\nabla_{t,x}\phi_{k+1}(t,x) + \frac{1}{\tau}m_k(t,x)$ in the convex set K_q. If we have a method for this projection in $\mathbb{R}^{n+1}$, requiring a constant number of operations, then the cost for this pointwise step is $O(N)$.

- Finally we update m by setting $m_{k+1} = m_k - \tau(\xi_{k+1} - \nabla_{t,x}\phi_{k+1})$.

This algorithm globally requires $O(N\log N)$ operations at each iterations (warning: N is given by the discretization in space-time). It can be proven to converge, even if, for convergence issues, the reader should rather refer to the works by K Guittet and Benamou-Brenier-Guittet, [189, 37, 190], as the original paper by Benamou and Brenier does not insist on this aspect. Also see the recent paper [195] which fixes some points in the proof by Guittet.

Compared to other algorithms, both those that we will present in the rest of the chapter and other more recent ones, the Benamou-Brenier method has some important advantages:

1. it is almost the only one for the moment which takes care with no difficulties of vanishing densities, and does not require special assumptions on their supports;

2. It is not specific to the quadratic cost: we presented it for power costs $c(x,y) = |x-y|^p$ but can be adapted to other convex costs $h(x-y)$ and to all costs issued from a Lagrangian action (see Chapter 7 in [293]); it can incorporate costs depending on the position and of time, and is suitable for Riemaniann manifolds;

3. it can be possibly adapted to take into account convex constraints on the density ρ (for instance lower or upper bounds);

4. it can handle different "dynamical" problems, where penalization on the density or on the velocity are added, as it happens in the case of mean-field games (see Section 8.4.4) and has been exploited, for instance, in [39];

5. it can handle multiple populations, with possibly interacting behaviors.

We refer for instance to [97, 248] for numerical treatments of the variants of the points 2. and 3. above, and to [38] for an implementation of the multi-population case. Here we present a simple picture (Figure 6.1), obtained in the easiest case via the Benamou-Brenier method on a torus.

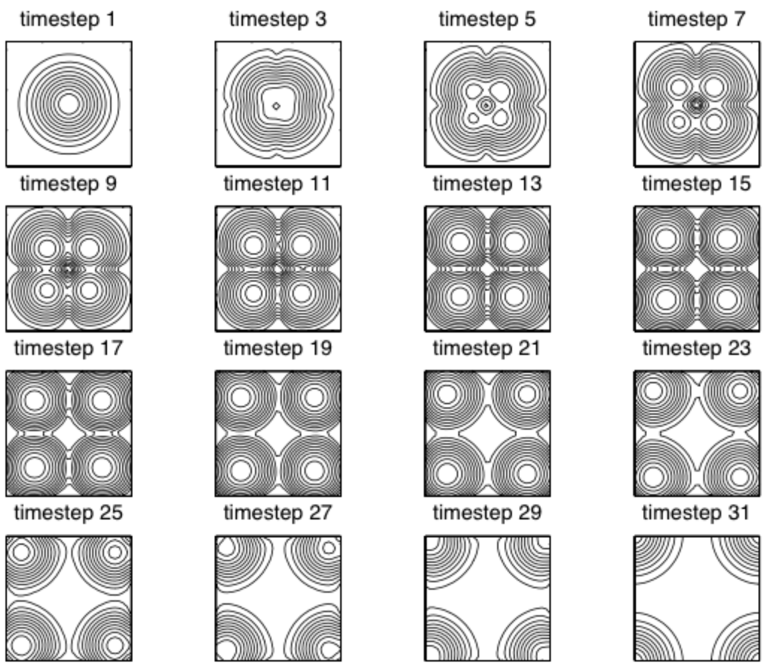

Figure 6.1: Geodesics between a gaussian on a 2D torus, and the same gaussian displaced of a vector $(1/2,1/2)$. Notice that, to avoid the cutlocus (see Section 1.3.2), the initial density breaks into four pieces. Picture taken from [36] with permission.

6.2 Angenent-Haker-Tannenbaum

The algorithm that we see in this section comes from an interesting minimizing flow proposed and studied in [20]. The main idea is to start from an admissible transport map T between a given density f (that we will suppose smooth and strictly positive on a nice compact domain Ω) and a target measure ν (we do not impose any kind of regularity on ν for the moment), and then to rearrange the values of T slowly in time, by composing it with the flow of a vector field $\mathbf{v}$, i.e. considering $\mathrm{T}\circ(Y_t)^{-1}$ instead of

$\mathrm{T}(x)$, where $Y_t(x) := y_x(t)$ is, as usual, the solution of

$$\begin{cases} y_x'(t) = \mathbf{v}_t(y_x(t)), \\ y_x(0) = x. \end{cases}$$

The vector fields $\mathbf{v}_t$ have to be chosen so that the map Y_t, defined via $Y_t(x) := y_x(t)$ preserves the measure f, and also so as to reduce the value of the Monge cost

$$M(\mathrm{T}) = \int c(x, \mathrm{T}(x)) f(x)\,\mathrm{d}x.$$

The cost c is taken smooth (say, C^1, we will see later the particular case of the quadratic cost).

The first computation to do is the derivative of the cost function: set $\mathrm{T}_t(x) := T(y_x(-t))$ where y is defined as the flow of an autonomous vector field $\mathbf{v}$. The condition for $\mathbf{v}$ to preserve the measure f is that the solution of

$$\partial_t \rho_t + \nabla \cdot (\rho_t \mathbf{v}) = 0 \quad \text{with } \rho_0 = f$$

is the constant (in-time) density $\rho_t = f$. This requires $\nabla \cdot (f\mathbf{v}) = 0$. We will set $\mathbf{w} = f\mathbf{v}$ for simplicity. By choosing $\mathbf{v}$ satisfying this condition we can write

$$\frac{d}{dt}\left(\int_\Omega c(x, \mathrm{T}(y_x(-t))) f(x)\,\mathrm{d}x\right) = \frac{d}{dt}\left(\int_\Omega c(y_x(t), \mathrm{T}(x)) f(x)\,\mathrm{d}x\right)$$
$$= \int_\Omega \nabla_x c(y_x(t), \mathrm{T}(x)) \cdot y_x'(t) f(x)\,\mathrm{d}x = \int_\Omega \nabla_x c(y_x(t), \mathrm{T}(x)) \cdot \mathbf{v}(y_x(t)) f(x)\,\mathrm{d}x,$$

and, at $t = 0$, this gives a derivative equal to $\int_\Omega \nabla_x c(x, \mathrm{T}(x)) \cdot \mathbf{v}(x) f(x)\,\mathrm{d}x$. In the computation above, we used the fact that Y_{-t} preserves the measure f, and that its inverse is Y_t (which is true for vector fields $\mathbf{v}$ independent of time).

We recover here an important part of the theory of optimal transport: if T is optimal, then the vector field $\nabla_x c(x, \mathrm{T}(x))$ must be orthogonal to all divergence-free vector fields $\mathbf{w}$, and hence it is a gradient (see Proposition 1.15). In the particular case $c(x,y) = \frac{1}{2}|x-y|^2$, this gives $x - \mathrm{T}(x) = \nabla\varphi(x)$ and hence T itself is a gradient. The fact that T is the gradient of a convex function, and not of any function, comes from second-order optimality conditions, at least when one has sufficient regularity (see **Ex**(44)).

On the other hand, the goal here is not to study the structure of an optimal T, but to start from a non-optimal T and to improve it. In order to do this, we must choose a suitable vector field $\mathbf{v}$. This will be done by producing a flow T_t, and the vector field $\mathbf{v}$ will be chosen in some clever way at every instant t of time. Hence, it will be no more autonomous. Several choices for $\mathbf{v}$ are possible, but we start from the simplest one: we saw that the derivative of the cost M is given by $\int_\Omega \xi \cdot \mathbf{w}$, where $\xi(x) = \nabla_x c(x, \mathrm{T}(x))$ and $\mathbf{w} = f\mathbf{v}$. Since $\int_\Omega \xi \cdot \mathbf{w}$ is a scalar product in L^2, this identifies the gradient of M for the L^2 structure, when restricted to the set of infinitesimal displacements preserving the measure (i.e. we impose the constraint $\nabla \cdot \mathbf{w} = 0$). The optimal choice of $\mathbf{w}$ (if we want to have a bound on $||\mathbf{w}||_{L^2}$) is to take as $\mathbf{w}$ the projection of $-\xi$ onto the vector space of divergence-free vector fields. Concerning the boundary conditions to associate to this divergence constraint, we choose as usual the Neumann conditions $\mathbf{w}\cdot\mathbf{n}=0$, which are also those required to preserve the measure without letting any mass exit Ω.

Box 6.2. **– Memo** – Helmholtz decomposition

Given a vector field $\xi\in L^2(\Omega;\mathbb{R}^d)$ it is always possible to write it as the sum of a gradient and of a divergence-free vector field $\xi=\nabla u+\mathbf{w}$ with $u\in H^1(\Omega)$ and $\nabla\cdot\mathbf{w}=0$ (in the distributional see). The decomposition is unique if one imposes $u\in H^1_0(\Omega)$, or $\mathbf{w}\cdot\mathbf{n}=0$ (in the weak sense, which, together with the zero-divergence condition, means $\int\mathbf{w}\cdot\nabla\phi=0$ for all $\phi\in C^1(\Omega)$).

These decompositions have a variational origin: indeed one can solve

$$\min\left\{\int|\xi-\nabla u|^2\,:\,u\in H^1_0(\Omega)\right\}$$

and get the first one, or

$$\min\left\{\int|\xi-\nabla u|^2\,:\,u\in H^1(\Omega),\int u=0\right\}$$

and get the second one (if Ω is connected). In both cases, the zero-divergence condition on $\mathbf{w}$ comes from the optimality conditions on u.

Note that ∇u is the projection of ξ on the set of L^2 gradients (of H^1_0 functions, in the first case) and hence $\mathbf{w}$ is the projection onto the set of divergence-free vector fields (with Neumann conditions, in the second case).

The projection $\mathbf{w}=P[\xi]$ can be computed by writing $\nabla\cdot\xi=\nabla\cdot\mathbf{w}+\Delta u$, i.e. $\mathbf{w}=\xi-\nabla(\Delta^{-1}(\nabla\cdot\xi))$.

Hence, we look for a family of transport maps T_t, a flow Y_t, a vector field $\mathbf{w}_t$ satifying the following conditions:

- T_t is defined as $\mathrm{T}_t=\mathrm{T}\circ(Y_t)^{-1}$ at every instant t
- the map Y_t is the flow of the time-dependent vector field $\mathbf{v}_t$, defined through $f\mathbf{v}_t=\mathbf{w}_t$
- the vector field $\mathbf{w}_t=P[-\xi_t]$, where P is the projection onto divergence-free vector fields with $\mathbf{w}\cdot\mathbf{n}=0$ and $\xi_t=\nabla_x c(x,\mathrm{T}_t(x))$

We can write the PDE satisfied by T_t, which is the transport equation $\partial_t\mathrm{T}_t+\mathbf{v}_t\cdot\nabla\mathrm{T}_t=0$.

Box 6.3. **– Memo** – Linear transport equation

Given a smooth vector field $\mathbf{v}_t$ its flow $Y_t(x)=y_x(t)$ (with, as usual, $y_x'(t)=\mathbf{v}_t(y_x(t))$ and $y_x(0)=x$), and a smooth function g, we define $g_t=g\circ Y_t^{-1}$. Then, we have

$$\partial_t g_t+\mathbf{v}_t\cdot\nabla g_t=0.$$

Indeed, we can write $g_t(y_x(t)) = g(x)$ and differentiate in time, thus getting the above equation. As soon as $\mathbf{v}_t$ stays Lipschitz continuous, its weak formulation is valid for arbitrary $g \in L^1$ since we can take a sequence of smooth approximating functions $g_n \to g$, write $\int\int g_n \circ Y_t^{-1}\left(\partial_t \varphi + \mathbf{v}_t \cdot \nabla\varphi\right)$, use the change-of-variable $y = Y_t^{-1}(x)$, and pass to the limit.

This equation is called *Transport equation*. It is similar to the continuity equation, but it is not in conservative form, and the mass of g is not conserved in time. On the contrary, L^∞ bounds on g are conserved, since the values of g_t are the same as those of $g = g_0$. The two equations, transport and continuity equations, are somehow dual: if a vector field $\mathbf{v}$ is fixed, and we take smooth solutions ρ_t of $\partial_t\rho_t + \nabla\cdot(\rho_t\mathbf{v}_t) = 0$ and g_t of $\partial_t g_t + \mathbf{v}_t \cdot \nabla g_t = 0$, then we have

$$\frac{d}{dt}\left(\int g_t \,\mathrm{d}\rho_t\right) = \int (\partial_t g_t)\rho_t + \int g_t(\partial_t\rho_t) = -\int \rho_t\mathbf{v}_t\cdot\nabla g_t - \int g_t\nabla\cdot(\rho_t\mathbf{v}_t) = 0.$$

We now compute formally the derivative in time of $M(\mathrm{T}_t)$ with the above choices

$$\frac{d}{dt}M(\mathrm{T}_t) = \frac{d}{dt}\left(\int c(x,\mathrm{T}_t)f\right) = \int \nabla_y c(x,\mathrm{T}_t)(\partial_t\mathrm{T}_t)f = -\int \nabla_y c(x,\mathrm{T}_t)\cdot\nabla\mathrm{T}_t\cdot\mathbf{w}_t.$$

We go on using $\nabla[c(x,\mathrm{T}_t(x))] = \nabla_x c(x,\mathrm{T}_t(x)) + \nabla_y c(x,\mathrm{T}_t(x))\nabla\mathrm{T}_t(x)$, together with $\int\nabla[c(x,\mathrm{T}_t(x))]\cdot\mathbf{w}_t(x)\,\mathrm{d}x = 0$, which comes from $\nabla\cdot\mathbf{w}_t = 0$. Hence we get

$$\frac{d}{dt}M(\mathrm{T}_t) = \int \nabla_x c(x,\mathrm{T}_t)\cdot\mathbf{w}_t(x) = \int \xi_t\cdot\mathbf{w}_t = -\int \xi_t P[\xi_t] = -||P[\xi_t]||^2 \le 0.$$

This shows that the proposed flow makes $M(\mathrm{T}_t)$ decrease in time, and that the value of $M(\mathrm{T}_t)$ is stationary if and only if $P[\xi_t] = 0$, i.e. if ξ_t is a gradient.

[20] proves the local existence of the above flow, i.e.

$$\begin{cases} \partial_t\mathrm{T}_t + \mathbf{v}_t\cdot\nabla\mathrm{T}_t = 0, \\ \nabla\cdot(f\mathbf{v}_t) = 0,\ \mathbf{v}_t\cdot\mathbf{n} = 0, \\ -\nabla_x c(x,\mathrm{T}_t) = f\mathbf{v}_t + \nabla u_t, \\ \mathrm{T}_0 \text{ given}, \end{cases}$$

(the second and third line can be replaced by $\mathbf{v}_t = \xi_t - \nabla(\Delta^{-1}(\nabla\cdot\xi_t))$, for $\xi_t = \nabla_x c(x,\mathrm{T}_t)$, which avoids inserting a new variable u_t, but introduces nonlocal terms in the equation) is proven. Also, it is proven that the only limits of T_t for $t\to\infty$ can be those rearrangements T of T_0 such that $\nabla_x c(x,\mathrm{T})$ is a gradient. The same is studied also in the more general case of transport plans instead of transport maps, but we avoid presenting this advanced case for the sake of simplicity. However, we note that for costs c which do not satisfy the twist condition, the framework of transport plans cannot be avoided. Indeed, in general it is not possible to converge to an optimal transport map as $t\to\infty$, simply because such a map could not exist.

In the very same paper [20] alternative choices of the vector field $\mathbf{v}$ are proposed, in particular "regularized" flows (i.e. applying regularization kernels to the above $\mathbf{w}_t$, and in this case global existence is proven). There is also a possibility of a local choice for the flow: in coordinates, we can take $\mathbf{w}^j = \xi^i_{ij} - \xi^j_{ii}$ (superscripts are components and subscripts are derivations). It can be checked that this choice as well guarantees that $M(\mathrm{T}_t)$ decreases in time.

Even if, as we noted, the method is general for any cost, it is in the case of the quadratic cost that it has been mainly employed. The idea in this case can be roughly summarized as "start from an admissible transport map, and then *gradientize* it" (i.e. transform it slowly into a gradient).

The equation becomes, for $\xi_t = x - \mathrm{T}_t$,

$$\begin{cases} \partial_t \xi_t - \mathbf{v}_t + \mathbf{v}_t \cdot \nabla \xi_t = 0, \\ \mathbf{v}_t = \frac{1}{f}\left(\xi_t - \nabla(\Delta^{-1}(\nabla \cdot \xi_t))\right), \\ \xi_0(x) = x - \mathrm{T}_0(x) \text{ given,} \end{cases}$$

From an implementation point of view (see [21]), the above system has been discretized and solved using an upwind scheme for the transport equation (roughly speaking, a finite difference method where the choice of the left or right-centered derivatives are made according to the sign of the components of the advecting vector field $\mathbf{v}$), and standard solvers for the solution of the Laplacian (such as Matlab Poisson solver). The complexity of each time iteration is of the order of $N \log N$, N being the number of pixels in the space discretization.

The AHT flow has been used for numerical purposes, with some advantages and disadvantages. Even if it has been mainly used for the quadratic case, it could be used for many smooth costs, which makes it a very general approach. Yet, from the numerical point of view the implementation involves some difficulties, and it has also the disadvantage of requiring strictly positive densities. Also, its convergence to the optimal map is not at all proven. This question can be easily understood in the quadratic case. Beware that we do not mean here rigorous convergence results: the problem is that the flow is meant to converge to a critical point for the cost M, which means (for the quadratic case) a transport T which is a gradient map. Nothing guarantees that it is the gradient of a convex map!

A priori, even the gradient of a concave function (i.e. the worst possible map for the quadratic cost, see **Ex**(2)) could be recovered. Since the flow has the property that $M(\mathrm{T}_t)$ is strictly decreasing in time as long as one does not meet a critical point, this case can be easily excluded (the only possibility is starting from T_0 being the gradient of a concave function, which is easy to avoid). But any other gradient could be found at the limit.

These issues have been discussed in [87]. It happens that a good idea to improve the results and enhance the chances of converging to the good transport map, following [20, 21], is to select a good initial guess T_0. Several choices (specific for the quadratic case) are proposed (including the Dacorogna-Moser transport map, see Box 4.3 in Section 4.2.3), but, at least at a first sight, the most suitable one is the Knothe transport (see Section 2.3). Indeed, with this choice, the initial transport map has triangular Jacobian matrix, with positive eigenvalues. Let us assume (extra than the previous assumptions) that ν is absolutely continuous with density $g(x)$ bounded from above and below by positive constants. As we know that the map T_t satisfies $(\mathrm{T}_t)_\# f = g$ for every t, the value of $|\det(D\mathrm{T}_t)|$ can never be 0 (is is bounded from below by $\inf f/\sup g$). If everything is smooth, this means that the sign of the determinant is preserved and that no eigenvalue can ever be zero (this is also underlined in [87]). Unfortunately, the eigenvalues of $D\mathrm{T}_t$ are real at $t=0$ (if we choose the Knothe map), are real at $t=\infty$ (since, as a gradient, the Jacobian is symmetric) but nothing prevents the eigenvalues to become complex and to pass from $]0,+\infty[$ to $]-\infty,0[$ without passing through 0 during the evolution! (even if this does not seem likely to happen, because the choice of $\mathbf{v}$ exactly forces T_t to become closer to the set of gradients, and gradients do not like complex eigenvalues).

A simple case, where there is no ambiguity if everything is smooth, is the 2D case. We can give a straightforward proposition.

Proposition 6.3. *Suppose that $(\xi_t,\mathbf{v}_t)$ is a smooth solution of the AHT system in the quadratic case, with $\mathrm{T}_t(x)=x-\xi_t(x)$ being a transport map between two given smooth densities $f,g\in\mathscr{P}(\Omega)$), where $\Omega\subset\mathbb{R}^2$ is a two-dimensional smooth convex and bounded domain. Suppose that* $\det(D\mathrm{T}_0(x))$ *is positive for every x. Also suppose that $\mathrm{T}_t\to\mathrm{T}_\infty$ in $C^1(\Omega)$ as $t\to\infty$ and that the evolution is non-trivial (i.e. there exists t such that $\mathrm{T}_t\neq\mathrm{T}_0$). Then T_∞ is the gradient of a convex function and is thus optimal.*

Proof. From the above considerations we have

$$M(T_0)-M(\mathrm{T}_t)=\int_0^t ||P[\xi_t]||^2_{L^2}\,\mathrm{d}t. \tag{6.1}$$

This implies that $\int_0^\infty ||P[\xi_t]||^2_{L^2}\mathrm{d}t<\infty$ and, at least on a sequence $t_n\to\infty$, we have $P[\xi_{t_n}]\to 0$ in $L^2(\Omega)$. Hence $P[\xi_\infty]=0$ since $\xi_{t_n}\to\xi_\infty:=x-\mathrm{T}_\infty(x)$ in C^1, and hence ξ_∞ and T_∞ are gradients. We need to prove that T_∞ is the gradient of a convex function, i.e. that the eigenvalues of $D\mathrm{T}_\infty$, which is now a symmetric matrix, are both positive (remember that we are in $\mathbb{R}^2$).

We know that $t\mapsto\det(D\mathrm{T}_t(x))$ is continuous in time, positive at $t=0$, and cannot vanish. From C^1 convergence, this implies $\det(D\mathrm{T}_\infty(x))>0$ for every x. Hence, the eigenvalues of $D\mathrm{T}_\infty(x)$ have the same sign (for each fixed x).

Let us look now at $\mathrm{Tr}(D\mathrm{T}_\infty(x))$. This is a continuous function of x, and it cannot be zero. Indeed, if it vanished somewhere, at such a point the two eigenvalues would have different sign. Hence, the sign of this trace is constant in x. Note that this only works for $t=\infty$, as for $t\in]0,\infty[$ one could a priori have complex eigenvalues.

If we exclude the case where $\mathrm{Tr}(D\mathrm{T}_\infty(x))<0$ everywhere we have concluded. Yet, this case corresponds to T_∞ being the gradient of a concave function (remember again that in dimension two, positive determinant and negative trace imply negative-definite). And concave function are the worse possible transports for the quadratic case (see **Ex**(2)). This is impossible since $M(\mathrm{T}_\infty)<M(\mathrm{T}_0)$ (except if the curve $t\mapsto\mathrm{T}_t$ were constant), from (6.1). □ □

Note that the assumptions on the initial condition of the above theorem are satisfied both by the Knothe transport and by the Dacorogna-Moser transport map.

Unfortunately, the above proof is really two-dimensional as we characterized positive symmetric matrices as those which have positive trace and positive determinant.

Also, in practice it is almost impossible to guarantee the assumptions of the Theorem above and convergence can only be observed empirically.

Open Problem (convergence of AHT): justify, even assuming smoothness and very strong convergence, that the AHT flow converges to the optimal map, if initialized at Knothe and/or Dacorogna-Moser, in dimension higher than two, and prove a rigorous result with minimal assumption in 2D.

6.3 Numerical solution of Monge-Ampère

As we saw in last section, we can use for numerical purposes the fact that solving the quadratic optimal transport problem is equivalent to finding a transport map which is the gradient of a convex function. In the last section, an iterated method was produced where a sequence of transport maps T_n (it was a continuous flow, but in practice it is discretized), with the same image measure, were considered, and they were forced to converge to a gradient. Here the approach is somehow opposite: a sequence of gradient maps is considered, and the corresponding measures are forced to converge to the desired target.

More precisely, the idea contained in [218] is the following: consider the simplified case where the domain $\Omega = \mathbb{T}^d$ is a d-dimensional torus, the source measure ρ is a $C^{0,\alpha}$ density bounded from below and from above, and the target measure is the uniform measure with constant density 1; we look for a smooth function φ such that $D^2\varphi \leq \mathrm{I}$ (which is the condition for φ to be c-concave, with $c(x,y) = |[x-y]|^2$ on the torus, see Section 1.3.2) and

$$\det(\mathrm{I} - D^2\varphi) = \rho.$$

We stress the fact that, in this periodic setting, it is not possible to look for a convex function u satisfying $\det(D^2u) = \rho$. Actually, periodic convex functions do not exist (except constant ones)! The only possibility is to stick to the framework of c-concave functions, without passing to the convexity of $x \mapsto \frac{1}{2}|x|^2 - \varphi(x)$.

If we find such a function φ, then $\mathrm{T}(x) = x - \nabla\varphi(x)$ (in the periodic sense) will be the optimal map between ρ and $\mathscr{L}^d$. Note that the choice of a uniform target measure gives a non-negligible simplification: the correct equation with a generic target measure would involve a function of $x - \nabla\varphi(x)$ at the denominator in the right-hand side. This would make the following analysis much more involved.

The idea of Loeper and Rapetti in [218] is to solve the equation through an iterative scheme, based on a Newton method.

Box 6.4. **– Memo** – Newton's method to solve $F(x) = 0$

Suppose we seek solutions of the equation $F = 0$, where $F : X \to X$ is a smooth function from a vector space X into itself. Suppose a solution $\bar{x}$ exists, with $DF(\bar{x})$ which is an invertible linear operator. Given an initial point x_0, we can define a recursive sequence in this way: x_{k+1} is the only solution of the linearized equation $F(x_k) + DF(x_k)(x - x_k) = 0$. In other words, $x_{k+1} = x_k - DF(x_k)^{-1} \cdot F(x_k)$. This amounts at iterating the map $x \mapsto G(x) := x - DF(x)^{-1} \cdot F(x)$. This map is well-defined, and it is a contraction in a neighborhood of